# ASSIGNMENT–1

## SOLID STATE PHYSICS

*Number of Questions: 6 | Marks per Question: 10*

### General Instructions

• There are 6 questions in the uploaded PDF. Each group (Max 10 people) must submit a report on two questions as the assignment. The assigned questions for the students' group are written in the Google Sheet.

• There will be a class presentation for each group with their answers for 15 minutes per group. Every student in the group should contribute to the assignment and mention who contributed which part.

• Prepare a detailed report for the assigned question.

• Mention the name and roll number of all participants involved in the assignment.

• Explain each step of the procedure/workflow in detail, including the theoretical background wherever relevant.

• Wherever Python programming is involved, include the complete Python source code in .py format, appropriate comments marking each major step of the code, screenshots of the interactive interface, results obtained for different input parameters, and a clear explanation and interpretation of the results.

• All graphs, crystal structures, XRD patterns, calculated quantities, and other outputs must be properly labelled and interpreted.

• Mention all references used at the end of the report, including textbooks, research papers, websites, COD, YouTube tutorials and software documentation, wherever applicable.

• Bonus point for a clean and well-arranged report.

## Question 1. Crystal Structure and XRD Analysis of Hydroxyapatite — 10 Marks

*Draw and analyse the crystal structure of Hydroxyapatite using experimental XRD data, PROFEX and VESTA.*

**1.** Download the provided .xrdml file from Google Drive and briefly describe the experimental XRD data.

**2.** Import the .xrdml file into PROFEX and perform the required Rietveld refinement using the phase apatite-OH and the instrument configuration cubix-ads-10 mm.

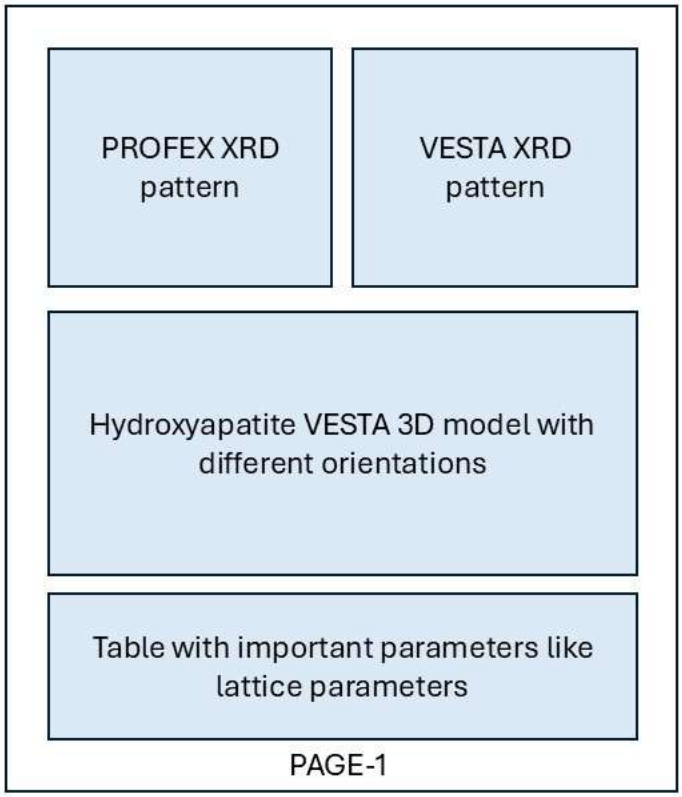
**3.** Analyse the refined XRD pattern and discuss the major diffraction peaks, peak positions, intensities, and the agreement between the experimental and calculated patterns.

**4.** Export the CIF file from the LST file obtained after refinement and explain the purpose of the CIF file in structural analysis.

**5.** Import the CIF file into VESTA and generate a three-dimensional model of the Hydroxyapatite crystal structure, clearly showing the constituent atoms and their arrangement.

**6.** Generate the XRD pattern using the VESTA software, and compare that with XRD pattern of the PROFEX software.

**7.** Discuss the crystal structure of Hydroxyapatite, including its crystal system, lattice parameters, atomic arrangement, and any important structural features observed from the VESTA model.




**8.** Summarise the XRD and structural analysis and provide appropriate references.

**9.** The first page of the report should be like the image shown.

## Question 6. Interactive Simulation of Magnetic-Field Penetration in a Diamagnetic Material --- 10 Marks

*Develop an interactive Python-based simulation and visualization interface to study the penetration of an externally applied magnetic field into a diamagnetic/superconducting material using the London equation.*

1. Briefly explain the London theory and the physical significance of the London equation in describing magnetic-field penetration into a superconducting/diamagnetic material.
2. Implement the London equation numerically in Python to calculate the spatial distribution of magnetic field inside and outside the material.
3. Develop an interactive user interface in which the user can select or enter the following parameters:

    Temperature (T)

    Applied magnetic field ($B_0$)

    Shape and dimensions of the material

    London penetration depth (lambda)

    Other relevant material parameters, where applicable.
4. Allow the user to select different sample geometries such as a slab, cylinder or rectangular sample and simulate how the geometry affects magnetic-field penetration.
5. Provide a 2D visualization of the magnetic-field lines around and inside the material, along with a suitable colour map/contour plot showing the magnitude of the magnetic field.
6. Incorporate the temperature dependence of the penetration depth, and demonstrate how changing temperature modifies the magnetic-field penetration into the material.
7. Provide suitable numerical/graphical output such as magnetic-field intensity versus distance from the surface and compare the results for different temperatures and applied magnetic fields.
8. Interpret the simulated results and explain how temperature, applied magnetic field, sample geometry and penetration depth influence magnetic-field screening and penetration based on the London model.

The final interface should be interactive, user-friendly and capable of updating the simulation and visualization whenever the input parameters are changed.